



# PLANET 7

## LHS 1140 B

- Description:** Dense, water-rich rocky super-Earth orbiting in the habitable zone. Its high density and quiet star make it one of the best targets to search for an atmosphere.
- Host star:** LHS 1140, a small, cool and unusually quiet red dwarf, about a fifth of the mass of the Sun.
- System:** One of 2 known planets orbiting LHS 1140, and the outer of the two.
- Location:** Constellation Cetus, about 49 light-years from Earth. Far too faint for the naked eye (magnitude 14).
- Discovery:** In 2017 by the MEarth Project, led by Jason Dittmann et al.

## Datasheet

| EXOPLANET DATA                                                                      |                  |                                 |
|-------------------------------------------------------------------------------------|------------------|---------------------------------|
|    | TEMPERATURE      | -40 °C                          |
|   | PRESSURE         | Unknown                         |
|  | ATMOSPHERE       | N <sub>2</sub> /CO <sub>2</sub> |
|  | GRAVITY          | 1.87 g                          |
|  | PLANET RADIUS    |                                 |
|  | ORBITAL PERIOD   |                                 |
|  | SEMI-MAJOR AXIS  |                                 |
|  | TRANSIT DEPTH    |                                 |
|  | TRANSIT DURATION |                                 |

| HOST STAR DATA                                                                      |                    |                      |
|-------------------------------------------------------------------------------------|--------------------|----------------------|
|    | NAME               | LHS 1140             |
|    | SPECTRAL TYPE      | M4.5V                |
|  | TEMPERATURE        | 3096 K               |
|  | DISTANCE           | 49 ly                |
|  | MASS               | 0.184 M <sub>☉</sub> |
|  | STAR RADIUS        | 0.216 R <sub>☉</sub> |
|  | AGE                | 5.0 Gyr              |
|  | APPARENT MAGNITUDE | 14.15                |
|  | NUMBER OF PLANETS  | 2                    |

### SURVIVAL CHALLENGE

Very dim red starlight combined with unusually high surface gravity

### NOTES



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| NOTES |
|-------|
|       |

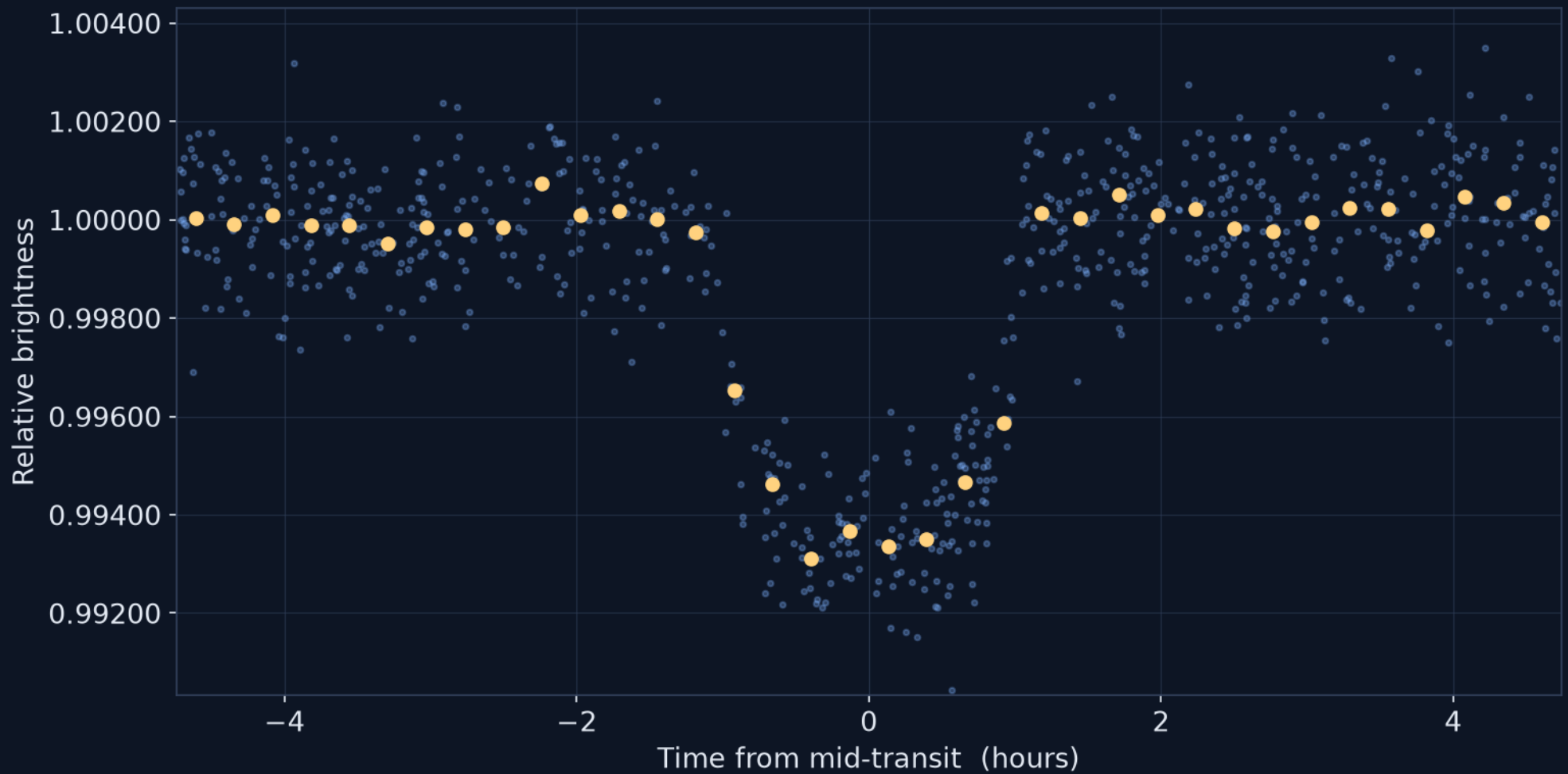
| PHYSICS                                                                                         |  |
|-------------------------------------------------------------------------------------------------|--|
|  PLANET RADIUS |  |
|  AI-ASSISTANCE |  |
|  REFLECTIONS   |  |

| MATHEMATICS                                                                                      |  |                                                                                                     |  |
|--------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------|--|
|  ORBITAL PERIOD |  |  SEMI-MAJOR AXIS |  |
|  AI-ASSISTANCE  |  |                                                                                                     |  |
|  REFLECTIONS    |  |                                                                                                     |  |

| CHEMISTRY                                                                                        |  |
|--------------------------------------------------------------------------------------------------|--|
|  SAMPLE DATA    |  |
|  AI-ASSISTANCE |  |
|  REFLECTIONS  |  |

| BIOLOGY                                                                                           |  |
|---------------------------------------------------------------------------------------------------|--|
|  ORGANISM      |  |
|  AI-ASSISTANCE |  |
|  REFLECTIONS   |  |

## LHS 1140 b: single transit



Blue points are simulated brightness measurements (many orbits phase-folded together, like real data); the gold points are binned averages.

LHS 1140 b: light curve over several orbits

